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Stereolithography in tissue engineering

Shelby A. Skoog · Peter L. Goering · Roger J. Narayan

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Abstract Several recent research efforts have focused on use of computer-aided additive fabrication technologies, commonly referred to as additive manufacturing, rapid prototyping, solid freeform fabrication, or three-dimensional printing technologies, to create structures for tissue engineering. For example, scaffolds for tissue engineering may be processed using rapid prototyping technologies, which serve as matrices for cell ingrowth, vascularization, as well as transport of nutrients and waste. Stereolithography is a photopolymerization-based rapid prototyping technology that involves computer-driven and spatially controlled irradiation of liquid resin. This technology enables structures with precise microscale features to be prepared directly from a computer model. In this review, use of stereolithography for processing trimethylene carbonate, polycaprolactone, and poly(D,L-lactide) poly(propylene fumarate)-based materials is considered. In addition, incorporation of bioceramic fillers for fabrication of bioceramic scaffolds is reviewed. Use of stereolithography for processing of patient-specific implantable scaffolds is also discussed. In addition, use of photopolymerization-based rapid prototyping technology, known as two-photon polymerization, for production of tissue engineering scaffolds with smaller features than conventional stereolithography technology is considered.

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S. A. Skoog · P. L. Goering
Division of Biology, Office of Science and Engineering Laboratories, Center for Devices and Radiological Health, U.S. Food and Drug Administration, Silver Spring, MD 20993, USA

S. A. Skoog · R. J. Narayan (✉)
Joint Department of Biomedical Engineering, University of North Carolina – Chapel Hill and North Carolina State University, Raleigh, NC 27965, USA
e-mail: roger_narayan@msn.com

1 Introduction

Stereolithography has become a valuable tool for fabrication of biocompatible tissue engineering scaffolds due to its ability to fabricate precise internal architectures and external geometries, which match those of human tissue. This rapid prototyping technique produces complex structures using a layer-by-layer approach that is based on spatially controlled solidification of liquid-based resins by photopolymerization. During fabrication by stereolithography, ultraviolet (UV) light is irradiated on the photosensitive resin surface in a precise pattern, which is defined by computer-aided design (CAD) files. Free radicals and other reactive species are formed upon excitation of photoinitiator molecules by UV light, causing polymerization of the resin and formation of a solid phase of the material. The first layer of photopolymerized polymer is adhered to a build platform, which provides support for structures as they are fabricated. Once this layer is polymerized, the platform is moved a defined step height for polymerization of the subsequent layer. The process of moving the support platform and curing each layer individually is repeated until the three-dimensional structure is complete.

Two different methods of irradiation may be applied to stereolithography, laser-based stereolithography and digital light projection stereolithography. The laser-based method is a direct write approach in which a computer-manipulated laser beam fabricates structures in a vector-by-vector,

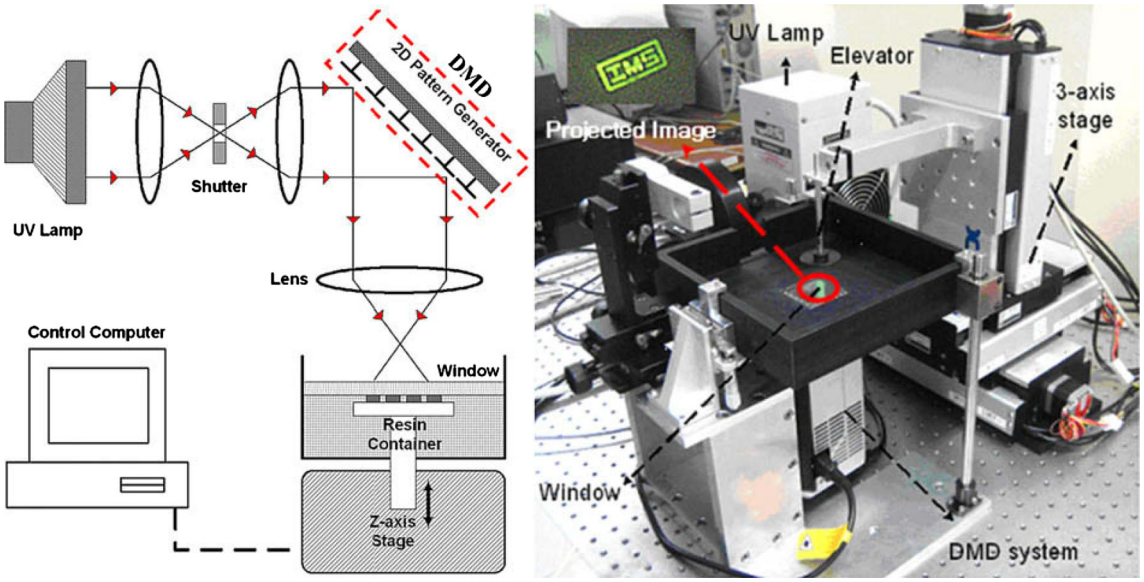


Fig. 1 Schematic drawing (*left*) and photograph (*right*) of the microstereolithography apparatus. Reprinted from [20]. Copyright (2009), with permission from Elsevier

bottom-up manner. In digital light projection, the UV light source is projected on a transparent surface at the bottom of a vat, which holds the photosensitive resin; an entire layer of material is simultaneously polymerized upon light exposure. In initial attempts involving this approach, a physical mask was applied to define the specific pattern to be illuminated during light projection stereolithography [1]. Resolution and design flexibility of projection stereolithography was greatly enhanced with the application of a liquid crystal display (LCD) to prepare specific irradiation patterns [2]. The projection system further evolved when digital micromirror devices (DMD) were applied as a dynamic pattern generator, which enabled high resolution stereolithography. DMDs employ several microscale mirrors that are arranged in an array, in which each mirror is capable of being independently rotated. An illustration and a photograph of a stereolithography apparatus (SLA) employing a DMD pattern generator is shown in Fig. 1. LCD and DMD pattern generators have significantly enhanced the efficiency and resolution of projection stereolithography.

Stereolithography enables significant freedom of design and is capable of fabricating minimum features sizes on the micrometer scale; although some stereolithography systems are capable of preparing structures with $\leq$5 μm features, most commercial systems prepare structures with $\geq$50 μm features [3]. The versatility in design and precise nature of stereolithography enables fabrication of complex scaffolds with physiologically relevant microstructures. Scaffolds with well-defined pore sizes, porosities, pore distributions, pore interconnectivity, and pore gradients have been fabricated using stereolithography. These scaffolds exhibit structural and mechanical properties that mimic that of native tissue and may permit cellular ingrowth and vascularization as well as transport of nutrients and waste. The ability to fabricate patient-specific scaffold implants using clinical medical imaging modalities, including computed tomography (CT) and magnetic resonance imaging (MRI), has further enhanced the utility of stereolithography in scaffold production for tissue regeneration. Advancements in resin materials as well as incorporation of bioactive materials and fillers have also improved the applicability of stereolithography in tissue engineering. This review focuses on stereolithography of novel materials, which exhibit a broad range of mechanical properties and biological functionalities.

2 Biodegradable materials processed using stereolithography

Fabrication of biodegradable structures using stereolithography was initially reported by Matsuda et al. [4]. In their work, photocurable liquid biodegradable copolymers of trimethylene carbonate (TMC) and poly(ε-caprolactone) (PCL) and were used to fabricate microscale surface features for biomedical applications. Significant research efforts have been dedicated over the past decade toward the development of stereolithography resins for biodegradable tissue engineering structures and other medical applications. Most polymers used in stereolithography for tissue engineering are functionalized with ester or carbonate bonds on the polymer backbone, which permits hydrolysis and facilitates biodegradability. Commonly examined

polymers for stereolithography of biodegradable structures include those containing poly(propylene fumarate) (PPF), trimethylene carbonate (TMC), poly(ε-caprolactone) (PCL), and poly(D,L-lactide) (PDLLA). Most resins result in fabrication of rigid networks applicable for bone tissue engineering or other hard tissues. It should be noted that recent research efforts have focused on developing elastic networks for soft tissue engineering applications.

3 Poly(propylene fumarate)-based biodegradable scaffolds

Poly(propylene fumarate) is one of the most widely studied polymers used in stereolithography that exhibits both biodegradability and photocrosslinking functionality. PPF is a linear polyester with an unsaturated backbone, which facilitates crosslinking with various agents such as diethylfumarate (DEF), methyl methacrylate, and N-vinylpyrrolidone (NVP) to reduce viscosity and to control other material properties. PPF is a biocompatible material for tissue engineering as it degrades into propylene glycol and fumaric acid subunits. Both degradation products are nontoxic; fumaric acid is a natural intermediate product in glucose metabolism. Furthermore, the mechanical properties of photocrosslinked PPF are similar to those of trabecular bone, making it a suitable biomaterial for skeletal tissue engineering.

In 2002, Fisher et al. [5] fabricated PPF scaffolds using a salt leaching fabrication technique. Their work demonstrated that the porous scaffolds elicited a bone healing response when implanted in a rabbit model. These scaffolds exhibited appropriate mechanical and biological properties for bone tissue engineering; however, the salt-leaching fabrication process provided little control over scaffold pore structure [5]. Stereolithography has been utilized for fabrication of PPF/DEF scaffolds with well-defined pore size, pore shape, and porosity. Cooke et al. [6] was first to fabricate PPF-based tissue engineering scaffolds using stereolithography by combining PPF with a non-reactive diluent, DEF, and a photoinitiator, bisacylphosphine oxide (BAPO). A SLA 250/40 (3D Systems, Rock Hill, SC, USA) instrument with a 325 nm wavelength laser and a custom build tank was used for scaffold fabrication. Solid cylindrical protrusions, holes, and slots were fabricated in the structure, which exhibited sharp edges and well defined features. This study demonstrated the feasibility of controlling the external and internal morphologies of a PPF scaffold using stereolithography [6]. Further studies to assess the cure depth of PPF-based resins have been conducted in order to optimize scaffold feature size [7, 8]. In a study by Choi et al. [7], cure depth experiments were used to determine appropriate fabrication parameters for creating scaffolds with well-controlled microstructures and geometries, including a PPF/DEF cubic scaffold and kidney scaffold with 100 μm interconnecting pores that was constructed from a CT-based human kidney model. The scaffolds were prepared using a PPF/DEF ratio of 7:3 (viscosity 150 cP at 50 °C) and a 2 % BAPO photoinitiator concentration. Fabrication of some horizontal channels on the kidney model failed, which may be attributed to viscous uncured resin remaining on the structure [7]. Recently, Brandi et al. demonstrated controllable vertical and horizontal resolution on the sub-micrometer scale for single layer PPF/DEF structures [8]. Curing depth optimization enabled a minimum layer thickness of approximately 300 nm. These results show significant promise for high resolution scaffold fabrication; however, the attainable aspect ratio is constrained by the resin absorption depth, which may be a limiting factor for fabrication of 3D tissue engineering scaffolds [8]. Future studies are needed to optimize processing parameters for creating 3D PPF scaffolds with small-scale features.

In each of these studies, different ratios of PPF/DEF blend were utilized; BAPO was used as the photoinitiator. Since the polymer resin composition has a significant impact on the photopolymerization process, the optimal PPF/DEF resin composition must be determined for high-resolution stereolithography. Multiple studies have examined various ratios of PPF/DEF and photoinitiator concentrations for scaffold fabrication [9, 10]. Most PPF/DEF resins applied for stereolithography of scaffolds have used weight ratios between 60:40 and 75:25 PPF/DEF and BAPO concentrations ranging from 0.5 to 2 % [6–10]. These resin combinations provide sufficiently low viscosities to permit fabrication by stereolithography, enabling production of scaffolds with appropriate mechanical properties for skeletal tissue engineering applications.

A significant advantage of using stereolithography to fabricate tissue engineering scaffolds is the control over the internal and external geometry of the structure. The ability to fabricate scaffolds with well-defined pore sizes, pore geometries, and porosities enables accurate evaluation of cell response to structural characteristics. This approach is particularly important in bone tissue engineering since natural bone itself is a porous tissue. Kim et al. [10] provides an extensive review of osteogenic differentiation and signal expression as a result of various bone scaffold designs that were fabricated using stereolithography. The effect of scaffold structure on osteoblastic response has been studied using PPF/DEF scaffolds with different pore sizes, porosities, and scaffold geometries. Lee et al. evaluated pre-osteoblastic response to scaffold pore size and scaffold structure (e.g., a "lattice" structure vs. "staggered" structure). The study demonstrated the importance of scaffold internal architecture; cells exhibited higher proliferation over 2 weeks on

staggered scaffolds than on lattice scaffolds with the same pore size and porosity. The results also demonstrated increased cell proliferation and enhanced mechanical properties (e.g., greater elastic modulus and ultimate strength) from stereolithography-fabricated scaffolds than from standard particulate leaching/gas foaming-fabricated scaffolds with the same porosity [11]. Kim et al. [12] further demonstrated the superiority of PPF/DEF scaffolds fabricated by stereolithography over scaffolds fabricated using a particulate leaching technique. Both scaffold types had a porosity of approximately 60 % with interconnected pores ranging from 200 to 800 μm. Stereolithography-based PPF/DEF scaffolds with controlled architectures were more permeable to cells and osteoblasts exhibited higher fibroblast growth factor (FGF-2), vascular endothelial growth factor (VEGF), and transforming growth factor beta 1 (TGF-B1) expression than osteoblasts on particulate leached scaffolds with random architectures [12].

Recent work has evaluated surface modification of PPF-based scaffolds that were created using stereolithography. The surface characteristics of tissue engineering scaffolds play a significant role in controlling initial cell responses to a biomaterial and may influence cell proliferation, adhesion, and differentiation. Arg–Gly–Asp (RGD) peptides [13, 14], bioactive ceramics [13, 15], and growth factors have all been incorporated into PPF/DEF scaffolds to enhance biological activity [16]. RGD-KRSR peptide sequences were noted to enhance initial cell attachment, proliferation, and differentiation of MC3T3-E1 pre-osteoblasts on PPF/DEF scaffolds [14]. Scaffolds with inorganic apatite coatings [13] and scaffolds with embedded hydroxyapatite nanoparticles [15] demonstrated increased MC3T3-E1 pre-osteoblast proliferation compared to corresponding control scaffolds. Recently, bone morphogenic protein 2 (BMP-2) loaded poly(lactic-co-glycolic acid) (PLGA) microspheres were incorporated within PPF/DEF scaffolds. The microspheres were suspended in the PPF/DEF photopolymer; scaffolds were fabricated out of this material using a stereolithography approach. Results involving implantation of scaffolds within cranial bone defects in Wistar rats demonstrated that gradual release of BMP-2 growth factor from the scaffold promoted bone formation [16].

4 Trimethylene carbonate, polycaprolactone, and poly(D,L-lactide) scaffolds fabricated using stereolithography

Stereolithography-fabricated scaffolds with a variety of polymeric resin materials and a broad range of mechanical properties have been fabricated for both soft and hard tissue engineering applications. Unlike PPF/DEF scaffolds, which are mechanically rigid, scaffolds fabricated with

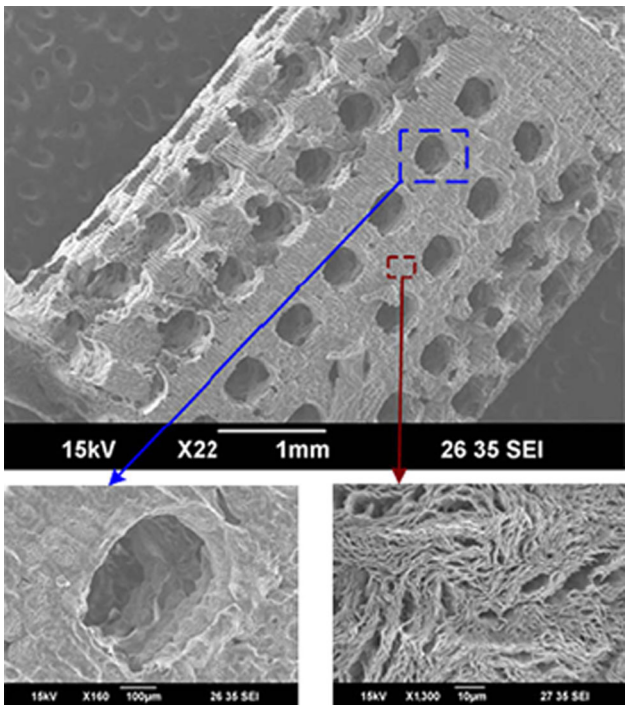


Fig. 2 Bi-pore scaffold fabricated using the molding process combined with phase inversion. Reprinted from [20], Copyright (2009), with permission from Elsevier

trimethylene carbonate (TMC)/trimethylolpropane (TMP) resin are more flexible. Lee et al. [17] has investigated TMC/TMP scaffolds for cartilage tissue engineering and demonstrated the significance of internal scaffold geometry on chondrocyte adhesion using in vitro studies. Scaffolds with a staggered lattice pattern showed greater cell adhesion and cell viability compared to scaffolds with a triangular pattern [18]. In a subsequent study, chondrocytes were encapsulated in an alginate hydrogel that was injected into a TMC/TMP framework in order to create a hybrid scaffold. In vitro analysis indicated that the scaffolds effectively maintained the phenotype of the encapsulated chondrocytes; normal lacunae formation was noted [18]. Schuller-Ravoo et al. [19]. recently optimized stereolithographic fabrication of TMC scaffolds by varying diluent concentrations, photocross-linking temperatures, and irradiation exposures. This approach enabled fabrication of high-resolution scaffolds with intricate (e.g., gyroid pore) network architectures, 53–66 % porosities, and pore sizes of approximately 300–400 μm. Successful structures were fabricated at 65 °C using TMC, up to 45 % propylene carbonate diluent, and molecular weights below 20,150 g/mol.

PCL is an elastic biodegradable polymer; complex tissue engineering scaffolds can be processed from this material using stereolithography. Indirect stereolithography and conventional scaffold fabrication methods have been integrated to fabricate PCL scaffolds, which contain large global pores with diameters of 300–400 μm and local pores with

diameters ranging from 30–100 to 1–5 μm [20]. Indirect stereolithography is an approach that involves processing a structure ("structure A") using stereolithography, which is subsequently used to make the usable structure ("structure B"); soft lithography methods (e.g., microcontact molding, microcontact printing, replica molding, or microtransfer molding) are commonly used to create "structure B." Stereolithography was used to prepare a sacrificial, soluble mold, in which the PCL resin was poured and set [20]. Salt leaching or phase inversion was used to form local pores. The salt porogens and mold were removed with NaOH, leaving the bi-pore scaffold as shown in Fig. 2 [20]. Elomaa et al. [21] recently developed photocrosslinkable PCL-based resins in which the PCL oligomers were functionalized with methacrylic anhydride. The photocrosslinked PCL networks were evaluated in an in vitro study involving NIH/3T3 fibroblasts. The cells proliferated on the PCL networks on day 1 after seeding and cell viability matched that of the control tissue culture polystyrene sample on day 3 after seeding. A Perfactory Mini Multilens SLA machine with DMD light projection (EnvisionTEC, Gladbeck, Germany) was used to fabricate cylindrical structures with a gyroid pore network as well as cuboidal structures with a diamond pore network. Since solvent dilution was not performed, no material shrinkage was noted and structures with precise geometries were obtained [21].

PDLLA has been applied for bone tissue engineering since it is a biodegradable polymer which exhibits mechanical properties approaching those of bone. Melchels et al. [22] and Jansen et al. [23] have extensively studied PDLLA-based resins for stereolithography of complex scaffolds with various architectures. Branched PDLLA oligomers were functionalized with methacryloyl chloride and were subsequently photocrosslinked; ethyl lactate was used as a non-reactive diluent. Altering the molecular weight of the polymer, which was associated with the arm length and the number of arms per macromolecule, permitted variation in crosslink density and influenced the mechanical properties of the structures. Mouse MC3T3 pre-osteoblast proliferation and spread morphologies were comparable for all of the materials [22]. Scaffolds with gyroid architectures were fabricated with fumaric acid monoethyl ester (FAME) functionalized PDLLA oligomers; N-vinyl-2-pyrollidone (NVP) was used as the diluent [23]. Mechanical properties, degradation profiles, and biocompatibility were all evaluated for various NVP concentrations [23]. Results showed that by altering the NVP concentration, water uptake and mechanical stiffness could be modified. In vitro studies demonstrated mouse pre-osteoblast adhesion and proliferation on scaffolds fabricated from the tested resin compositions [23]. In subsequent studies, PDLLA and P(DLLA-co–CL) resins were used to fabricate scaffolds of various geometries using 2 wt% ethyl-2,4,6-trimethylbenzoylphenylphosphinate (TPO-L) photoinitiator, 40 wt% N-methylpyrrolidone (NMP) non-reactive diluent, and 0.15–0.2 wt% Orasol orange G dye [24, 25]. These studies demonstrated the freedom of scaffold design associated with stereolithography of PDLLA-based resins; high-resolution scaffolds with cubic, diamond, and gyroid unit structures were fabricated [24, 25]. Furthermore, a gyroid scaffold with a pore size and porosity gradient was fabricated; the micro-CT image of this complex design is shown in Fig. 3 [25]. The mechanical properties of these scaffolds varied from rigid to flexible, enabling use of these scaffolds for a broad range of tissue engineering applications.

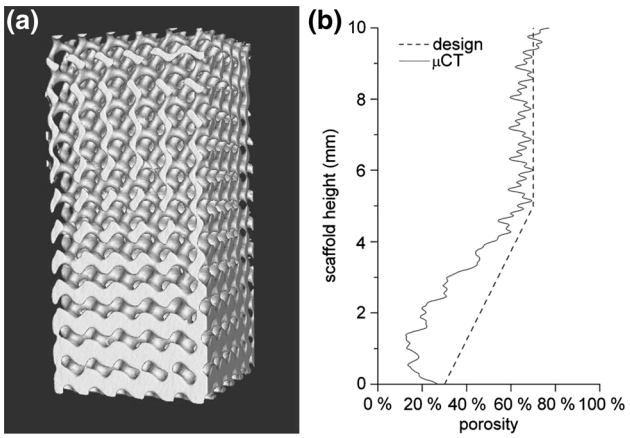


Fig. 3 Built PDLLA scaffold with gyroid architecture, showing a gradient in porosity and pore size. **a** uCT visualization. **b** Change in the average porosity with scaffold height (*solid line*) in comparison with the designed porosity (*dotted line*). Reprinted from [25], Copyright (2010), with permission from Elsevier

Natural polymers, including chitosan and gelatin, have also been fabricated into scaffold structures using stereolithography. For example, gelatin hydrolysate was modified with methacrylamide moieties and other moieties to control scaffold density and mechanical properties [26, 27]. Subsequent studies evaluated cytotoxicity of various methacrylate-based gelatin derivatives. In vitro studies using MC3T3-E1 osteoblast cells and human umbilical vascular endothelial cells (HUVEC) demonstrated the low cytotoxicity of the gelatin-based monomers [27]. Jiankang et al. [28] utilized a chitosan/gelatin resin to fabricate predefined scaffold architectures with indirect stereolithography. In this process, stereolithography was used to prepare a resin mold that was subsequently utilized to produce a polydimethylsiloxane (PDMS) mold. The hybrid gelatin-chitosan resin was subsequently cast into the mold and lyophilized. Scaffolds were fabricated to mimic the internal unit structure of the liver, the hepatic lobule, as shown in Fig. 4 [28, 29]. The scaffold architecture included accurate branching angles of liver vasculature and a hepatocyte chamber [28]. Hepatocyte–scaffold interactions were analyzed with in vitro studies over 3 days; the hepatocytes exhibited greater attachment and

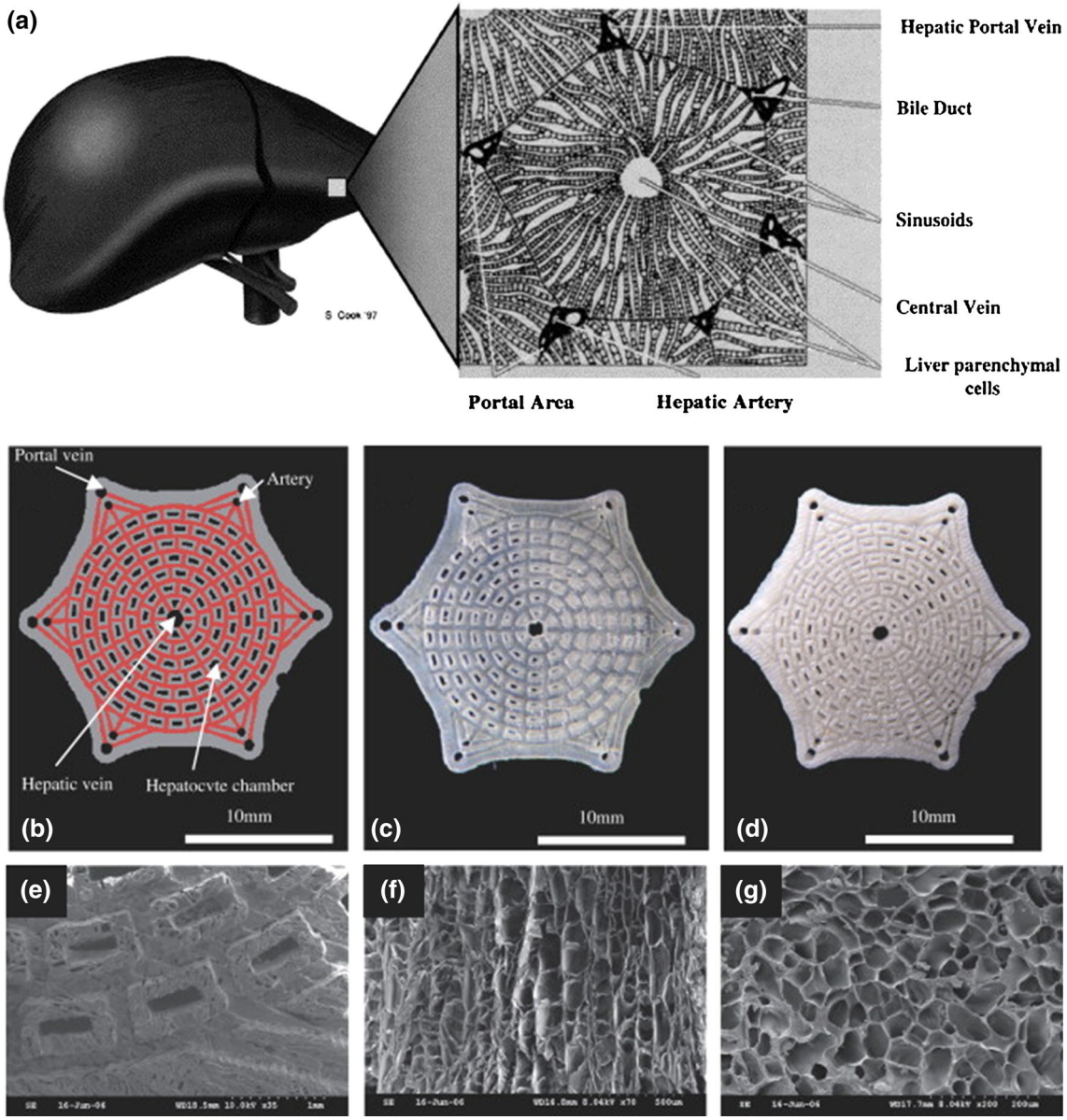


Fig. 4 **a** Liver and *inset* of basic functional unit, the hepatic lobule. The liver consists of four lobes of unequal size and shape. The portal vein subdivides repeatedly, terminating into capillaries leading to a lobule (see labeled histologic features). Liver tissue is composed of thousands of lobules that are primarily comprised of parenchymal cells, the metabolic cells of the liver. Reprinted from [29], Copyright (2005), with permission from Elsevier. Porous chitosan/gelatin scaffold with specific external shape and predefined internal morphology. **b** CAD model, **c** resin mold fabricated using SL technique, **d** porous chitosan/gelatin scaffold, **e** SEM of the predefined internal morphology, **f** microstructures when segmented longitudinally and **g** microstructures when segmented transversely. The device structure labeled "hepatic vein" in image **b** is representative of the anatomical structure labeled "central vein" in image **a**. Reprinted from [28], Copyright (2007), with permission from Elsevier

proliferation on scaffolds with well-defined branching structures than on corresponding control scaffolds [28].

5 Bioceramic fillers and bioceramic scaffold fabrication

Bioceramic materials have been incorporated within tissue engineering scaffolds using several fabrication techniques for enhanced bioactivity and improved mechanical properties. For example, bioceramic fillers have been incorporated into photocrosslinkable polymer resins for direct fabrication by means of stereolithography. Direct stereolithography of bioceramic composite scaffolds is difficult since the addition of inorganic components increases the resin viscosity; in addition, bioceramic particles may interfere with irradiation of the polymer. Due to limitations

on the bioceramic concentration via the direct stereolithography approach, a number of studies have utilized indirect stereolithography (e.g., fabrication of a negative mold using stereolithography). The mold is then filled with the bioceramic slurry, the complex undergoes pyrolysis to remove the mold, and the remaining bioceramic structure is sintered. This method permits a high concentration of bioceramic material and does not require a photopolymerizable ceramic resin; however, indirect stereolithography increases the number of fabrication steps and does not provide the same freedom of design as direct stereolithography.

Hydroxyapatite (HA) has been widely studied for skeletal tissue engineering as its chemical composition is similar to the inorganic component of natural bone. Direct and indirect stereolithography have been utilized for processing of HA-based bone scaffolds. Popov et al. [30] and Barry et al. [31] demonstrated fabrication of HA-based olygocarbonatedimethacrylate (OCM-2®) composite scaffolds using an LS-250 stereolithography instrument with a focused helium-cadmium (HeCd) laser (ILIT RAS, Shatura, Russia). Supercritical carbon dioxide ($scCO_2$) post-treatment was applied to remove residual monomers and provide interconnected microporosities for improved cellular activity. The scaffolds were implanted into the femoral tissue of white laboratory rats; the results demonstrated that HA incorporation encouraged osteogenesis and osteointegration within the scaffold. Both in vivo and in vitro analysis showed higher attachment of cells on the $scCO_2$-treated materials. Indirect stereolithography has been applied for fabrication of HA scaffolds with various pore sizes, porosities, and scaffold geometries. Chu et al. demonstrated the feasibility of fabricating two types of HA cylindrical scaffolds with 40 % porosity using indirect stereolithography [32, 33]. Scaffolds with orthogonal channels and scaffolds with radial channels were fabricated and implanted in Yucatan minipig mandibles for periods of time up to 9 weeks. Both scaffold designs experienced bone ingrowth, with 45 % ingrowth in the orthogonal design and 23 % ingrowth in the radial design at 9 weeks post-implantation. The study also showed formation of different bone geometries within the various scaffold internal architectures [33]. Li et al. [34] further demonstrated freedom of design enabled by indirect stereolithography by fabricating anatomically shaped femoral supracondylar HA scaffolds based on CT imaging data from a dog. Calcium phosphate cements (CPC) were used to fabricate HA scaffolds with channels between 300 and 800 μm in diameter.

Indirect stereolithography and gel casting has been further utilized for processing of bioceramic scaffolds composed of beta-tricalcium phosphate (β-TCP) and bioactive glass. Like HA, these materials encourage bone tissue formation. Padilla et al. [35] used a negative mold fabricated by stereolithography to cast a bioactive glass suspension. Slurries composed of varying compositions and different dispersants were evaluated. Heat treatment was used to remove the molds; scaffolds with 400–470 μm channels, interconnected pores, and 1.4 μm diameter smaller pores were obtained using this approach [35]. Li et al. [36] used indirect stereolithography and gel casting to fabricate β-TCP scaffolds with 300–550 μm diameter cylindrical pores and an overall porosity of 45 %. Bian et al. [37] used stereolithography to fabricate a mold, which was then filled with a β-TCP nanoparticle ceramic slurry. Pyrolysis and sintering were used to remove the organic material, leaving only a pure β-TCP scaffold with radial and longitudinal interconnecting channels (600–800 μm diameter). Recent studies have integrated bioceramic scaffolds with natural polymers to fabricate complex hybrid scaffolds for tissue engineering. For example, Sanchez-Salcedo et al. [38] fabricated HA/β-TCP/agarose macroporous scaffolds with various porosities (59–80 %), pore sizes (0.2 and 70 μm), and interconnected channels. Bian et al. [39] used histological analysis of natural articular cartilage from human knees to design β-TCP/collagen scaffolds with a cartilage phase and a bone phase for tissue engineering of the cartilage-bone interface. Indirect stereolithography and gel casting was used to create the osteochondral scaffolds with fully interconnected pores (700–900 μm). Interlocking structures with channels (200–500 μm diameter) were used as a transitional phase to bind the cartilage and bone phases together. In vivo evaluation of the ceramic scaffold component showed adequate cell adhesion and proliferation over a 7 day culture period [39].

6 Photopolymerizable hydrogels

Photocrosslinkable hydrogels have been examined for many applications, such as encapsulation of cells, production of microenvironments mimicking the extracellular matrix, and fabrication of scaffolds with patterned arrays of biological materials. These hydrogels may be rendered biodegradable and can be functionalized by covalently attaching bioactive materials such as extracellular matrix (ECM) components, adhesion ligands, and growth factors. Structured hydrogels that are fabricated using stereolithography may be used for tissue engineering of soft tissues. In addition, they may be used for evaluating cell behavior inside complex 3D microenvironments in order to obtain greater understanding of tissue and organ function. Hydrogels exhibit numerous advantages as biomaterials, including permeability to nutrients, oxygen, and water-soluble metabolites as well as elasticity comparable to soft tissues.

Loading cells within scaffolds during scaffold fabrication may enable higher cell concentrations and better cell distribution than direct seeding onto porous scaffolds. Poly(ethylene glycol) (PEG)- based hydrogels have been commonly utilized as materials for cell encapsulation studies. For example, Dhariwala et al. [40] first successfully encapsulated cells in poly(ethylene oxide) (PEO) and poly(ethylene glycol) dimethacrylate (PEG-DMA) hydrogels using stereolithography. A high density of Chinese hamster ovary (CHO) cells was encapsulated in ring-shaped hydrogel scaffolds, which were fabricated using stereolithography. The cells remained viable and proliferated within the scaffold, which exhibited an elastic modulus (1 kPa) comparable to soft tissue [40]. Since this study, human dermal fibroblasts [41–43], mouse fibroblasts [44], mesenchymal stem cells [45], embryonic liver cells [46], and chondrocytes have been successfully photo encapsulated in stereolithography-fabricated PEG-based hydrogel structures.

A cell encapsulation approach that relies on photopolymerization of hydrogels by stereolithography has been applied to processing of functionalized hydrogels for spatial localization of cells. Several studies have incorporated adhesive peptide sequences, such as arginine-glycine-aspartic acid (RGD), into hydrogels for enhanced cellular activity [42–44, 46, 47]. For example, Lee et al. [43] demonstrated guided cell migration using glycine-aspartic acid-serine-lysine (RGDSK)-patterned PEG-DA-based hydrogels. In their work, 3D patterns of fluorescently tagged RGDSK peptide sequences were spatially organized within a biodegradable collagenase-sensitive PEG hydrogel using photolithography. Human dermal fibroblast clusters embedded in the hydrogel migrated from the cell clusters into the RGDSK-patterned regions of the hydrogel [43].

Additional studies have demonstrated the feasibility of depositing multiple bioactive materials or cell types into distinct regions of hydrogel scaffolds. Mapili et al. [47] showed spatial organization of several types of materials within PEG-DMA scaffolds that were prepared using stereolithography. RGD peptide- and heparin sulfate-conjugated scaffolds were fabricated as well as scaffolds with layers containing fluorescently-labeled latex microparticles. For example, a multilayer scaffold with 400 μm pores was fabricated with individual layers of either cyanine 5 (Cy5)- or fluorescein isothiocyanate (FITC)-labeled microparticles [47]. Arcaute et al. fabricated multi-material scaffolds with fluorescently labeled dextran and fluorescently labeled bioactive PEG (PEG conjugated with arginine-glycine-aspartic acid-serine (RGDS)) in PEG-DMA and PEG-DA hydrogels [42]. To fabricate multi-material scaffolds, a customized mini-vat instrument was developed that enabled self-alignment during fabrication. This instrument permitted easy removal of the vat for resin exchange and enabled removal as well as rinsing of the built structure between build layers. Figure 5 shows the different sizes, geometries, and spacing of structures fabricated using this process [42]. Cell seeding experiments showed specific cell localization of human dermal fibroblasts in locations that were patterned with the bioactive PEG. Chan et al. [44] showed that fluorescently-labeled cells may also be used to demonstrate multi-material spatial organization within PEG-DA hydrogels that were fabricated with stereolithography. Two cell suspensions of NIH/3T3 cells were used, one was labeled with a green-colored fluorescent dye and the other was labeled with a red-colored fluorescent dye. Distinct layers of the different cell solutions in resin were patterned in a single crosshatch-shaped scaffold using stereolithography [44].

Initially, hydrogel structures fabricated using stereolithography exhibited simple geometries with only a few build layers. More recent studies have utilized the versatility of stereolithography to fabricate complex, 3D structures using hydrogel-based resins. Using a projection stereolithography instrument, Seck et al. [45] fabricated poly(ethylene glycol)/poly(D,L-lactide)-based nonporous hydrogel structures with disk and film geometries as well as porous hydrogel structures with a gyroid pore network. The scaffolds exhibited good pore interconnectivity and narrow pore size distribution, with pore sizes from 387 to 558 μm. In vitro studies over 5 days demonstrated proliferation of human mesenchymal stem cells on the scaffolds [45]. Gauvin et al. [48] fabricated collagen-based gelatin methacrylate (Gel-MA) hydrogels into hexagonal or wood pile porous structures using various polymer concentrations. The hydrogel composition was reinforced with calcium carbonate ($CaCO_3$) microparticles. Mechanical properties were evaluated to investigate the effect of pore architecture and polymer concentration on compressive elastic modulus. The high strain elastic modulus of the wood pile structure made using 10 % GelMA was statistically higher than that of the hexagonal structure made using 10 % Gel-MA, demonstrating the significance of scaffold structure on mechanical properties. Human umbilical vascular endothelial cells (HUVECs) remained viable up to 7 days when seeded on both the hexagonal and woodpile scaffolds. Three-dimensional confocal images of the HUVEC cells transfected with green fluorescent protein (GFP) on the hexagonal scaffolds are shown in Fig. 6. The cells maintained their normal phenotype, demonstrated by expression of platelet endothelial cell adhesion molecule 1 (PECAM-1 or CD31), endothelial cell markers, and the blood glycoprotein Von Willebrand Factor (vWF) [49].

7 Patient-specific implantable scaffolds

The ability to transfer medical imaging data (e.g., CT and/or MRI data) to stereolithography makes this rapid

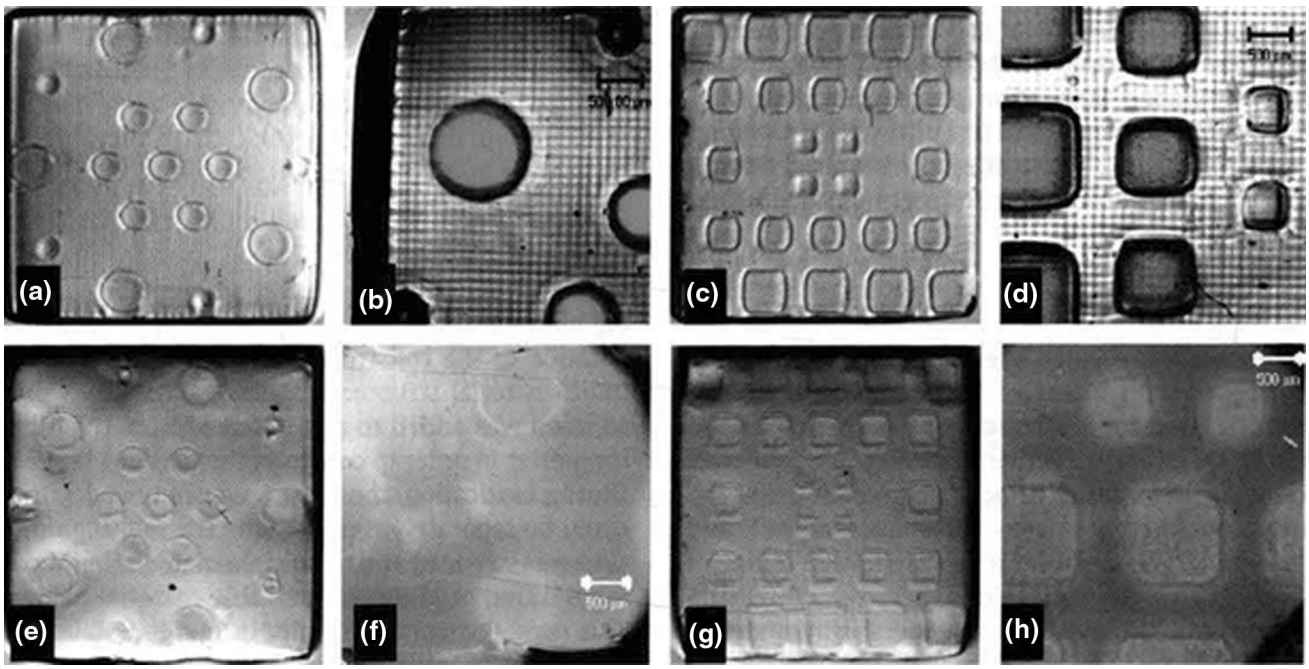


Fig. 5 Brightfield (**a** and **c** ×10; **e, g** ×7.1) and confocal microscope (×25) images of 2-D multi-material scaffolds fabricated using fluorescently labeled dextran. **a–d** Scaffolds fabricated with PEG-dma 1 K; **e–h** scaffolds fabricated with PEG-da 3.4 K as the main material. Reprinted from [42], Copyright (2010), with permission from Elsevier

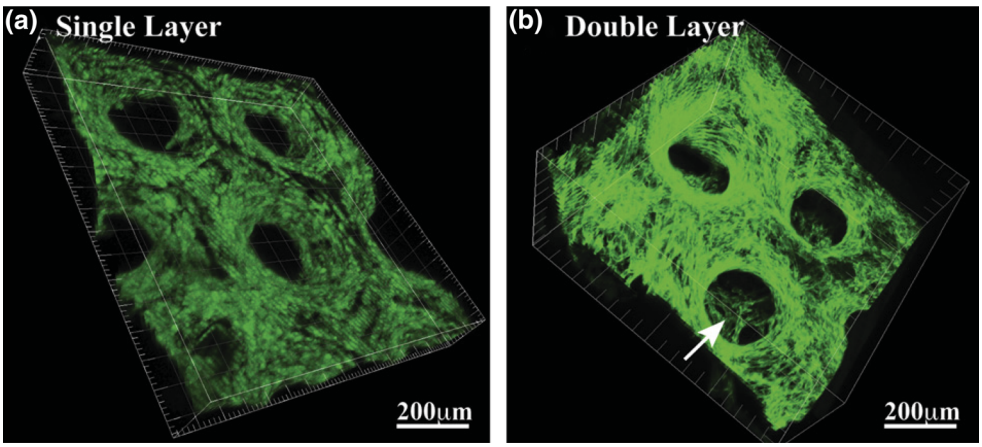


Fig. 6 3D confocal imaging of the HUVEC-GFP seeded scaffolds, showing the capability of the PSL system to generate **a** single and **b** multiple layer assemblies with precisely defined geometries. These images, taken 4 days following cell seeding, also demonstrate the efficiency of the seeding method and the structural stability of the scaffold, as the features initially engineered in the 3D structure remain present throughout the culture period. Reprinted from [48], Copyright (2012), with permission from Elsevier

prototyping technique valuable for fabrication of personalized, patient-specific devices. Stereolithography may be used to fabricate prostheses with geometries specific to a patient defect. Although direct fabrication of implants using stereolithography has not been fully realized, fabrication of several patient-specific implants using indirect stereolithography has been reported. Devices such as tissue engineered heart valves [49, 50], artificial mandibles [51], composite hemi-knee joints [52], and cranioplasty prostheses [53–55] have been prepared using stereolithography; these studies have utilized stereolithography-prepared molds and models to create custom implants. For example, Sodian et al. [50] fabricated a descending aorta and tailored tissue engineered heart valves which do not require suturing. In these studies, CT data and 3D reconstruction of the heart valve and vessel geometry were used to fabricate a stereolithographic plastic model. The model was utilized as a cast to prepare polymeric vascular scaffolds from polyglycolic acid (PGA), poly-4-hydroxybutyrate (P4HB), and polyhydroxyoctanoate (PHOH) [50].

Cranioplasty prostheses have been fabricated using stereolithography in order to avoid issues associated with improperly fitted implants. Stereolithography has been applied for fabrication of cranioplasty implants composed of carbon fiber reinforced polymers [53], bioceramics [54], and polymers [55]. A long-term evaluation of a porous hydroxyapatite cranioplasty prostheses fabricated using indirect stereolithography showed that these scaffolds are aesthetically adequate and experience minimal infection upon implantation [54]. CT data obtained 9 months following implantation of the scaffold showed initial fusion with adjacent skull bones, suggesting osteointegration. This study suggests that hydroxyapatite scaffolds with controlled porosity may be fabricated using stereolithography for tissue regeneration in cranial defects [54].

8 Two-photon polymerization of tissue engineering scaffolds

Two-photon polymerization (2PP) is a stereolithography approach in which the photosensitive resin is polymerized upon nearly simultaneous absorption of two photons using a femtosecond laser [56]. The confined energy causes a localized chemical reaction, which enables fabrication of features as small as 100 nm. Since the 2PP approach cures a three-dimensional unit volume (voxel) of material, 2PP does not involve movement of a platform in order to create a three-dimensional structure [56]. Instead, the laser focus is commonly moved along a two-dimensional scanning path; beam spot translation in the *Z*-axis allows three-dimensional structures to be created. The development of two-photon polymerization has permitted fabrication of scaffolds with nanoscale features similar to those within the natural extracellular matrix of natural tissue. The small-scale features obtained with 2PP enables further evaluation of the effects of scaffold structure and morphology on cell behavior and biological mechanisms. Additional details on the use of two photon polymerization to create scaffolds for tissue engineering are described in a recent review by Gittard et al. [57].

Ovsianikov et al. utilized 2PP to fabricate methacrylamide-modified gelatin (GelMOD) scaffolds with 1.5 μm topographic features [58]. The scaffolds contained 250 μm × 250 μm rectangular pores. These pores were filled with a mesh-like polymeric network, possibly due to diffusion-driven polymerization. In vitro cell studies using porcine mesenchymal stem cells demonstrated that this network functions to entrap cells and promote cellular integration within the scaffold; treatment in collagenase solution for up to 2 h resulted in partial or complete degradation of the mesh, causing poor cell adhesion upon seeding [58].

Psycharakis et al. [59] evaluated 2PP-fabricated scaffolds with various organic–inorganic compositions, geometries, and pore sizes. Hybrid materials of methacyrloxypropyl trimethoxysilane (MAPTMS) and methacrylic acid (MAA) with various ratios of zirconium n-propide, titanium, or alkoxysilane groups were fabricated into cubic and porous scaffolds with 2PP; a Ti sapphire femtosecond laser operating at 800 nm was used in this study. NIH/3T3 mouse fibroblasts were seeded up to 7 days on the scaffolds. The results indicated that that the organic–inorganic ratio of the scaffold material and the scaffold features affected cell growth; an increase in cell proliferation was associated with decreasing MAA content. Cellular attachment was not significantly different between cubic scaffolds of various sizes. The woodpile porous scaffolds had highest cell densities and the rectangular porous scaffolds with 20 μm pore sides had the lowest cell densities [59].

Recently, Kapyla et al. [60] used 2PP to fabricate unique scaffolds with hollow sphere repeating units for geometrical analysis of scaffold precision and cellular response. The commercial polymer-ceramic material Ormocomp® was used to make scaffolds of various pore diameters and unit cell dimensions. Scaffolds with the highest interconnectivity contained 60 μm diameter hollow spheres and 54 μm pore openings and were used for in vitro analysis involving human adipose stem cells. Live/dead staining demonstrated that the majority of cells remained viable for the entire culture duration; cells expressed normal mesenchymal stem cell biomarkers such as CD73, CD90, and CD105 [60].

9 Conclusions

Stereolithography is a rapid prototyping technique that permits fabrication of intricate designs with exceptional accuracy. This approach has been used to fabricate scaffolds with well-defined pore sizes, porosities, pore distributions, pore interconnectivity, and pore gradients out of a variety of biocompatible materials. Optimization of stereolithography processing parameters, novel photopolymerizable resins, and developments in the processing equipment have allowed precise fabrication of structures on the micrometer and submicrometer scale. The high resolution and freedom of design provided by stereolithography hold significant potential in biomedical engineering. Recently developed biodegradable and biocompatible materials exhibit mechanical, chemical, and biological characteristics that are more suitable for use in implants than previous materials. The control over feature size provided by stereolithography enables production of scaffolds with narrow variations in pore geometry and pore sizes; such structures may be used to evaluate the effects of scaffold structure on cellular response and tissue integration. Fillers and bioactive compounds may be used to enhance the mechanical integrity and biological functionality of

stereolithography-fabricated scaffolds, respectively. Stereolithography combined with a medical imaging modality (e.g., MRI or CT) enables advances in personalized medicine through rapid fabrication of patient-specific tissue engineering scaffolds. Numerous studies have shown the potential of these structures and materials as implants through in vitro analysis and in vivo animal studies; however, very few scaffolds fabricated by stereolithography have been used in clinical trials involving human subjects. Although the materials and procedures discussed in this review hold significant promise for tissue engineering scaffold fabrication, additional studies such as long-term toxicological studies and long-term mechanical studies are necessary in order to facilitate translation and widespread clinical use.

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